

Vladimir LAPIN, Ph.D.

Applied Scientist | Numerical Optimization & Forecasting | Operations & Workflow Systems

PERSONAL DATA

NATIONALITY: Russian | Authorized to work in the EU (Permanent Resident – Spain)
LOCATION: Barcelona Supercomputing Center, Barcelona, Spain
CONTACT: vladimir.lapin@bsc.es | +34 611 474 420
PROFILES: [Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#)

PROFILE SUMMARY

- **Applied Scientist / Numerical Optimizer:** Applied Mathematician (1st Honours, MSU) with 15 years designing large-scale solvers, probabilistic forecasting systems, and HPC pipelines—from formulation through scalable implementation to operational validation.
- **Capacity & Workflow Orchestration:** Architect of automated execution DAGs (Autosubmit/Slurm) scheduling 100s of concurrent compute jobs under capacity and deadline constraints for WMO/CMIP7 operational cycles; optimizing cluster throughput and resource allocation.
- **AI Adoption & Pipeline Validation:** Leading group-wide AI adoption and agentic workflows (Climate Commons); architect of schema-driven validation contracts, automated decision-support tools, and CI harnesses bridging mathematical rigor with production reliability.

SKILLS

MATH	Numerical optimization, capacity & queue scheduling, probabilistic forecasting, statistical anomaly detection, uncertainty quantification, root-cause diagnostics
CODE	Python (numpy, pandas, scipy.optimize, xarray, statsmodels), SQL (DuckDB, PostgreSQL), GIT, C/C++, FORTRAN (MPI)
DATA	Automated ETL & data pipelines, schema-driven validation harnesses, reproducible data versioning, multidimensional datasets (PARQUET, NETCDF/HDF5)
INFRA	Workflow Orchestration (AUTOSUBMIT, DAG scheduling, dependency tracking), Linux/BASH, HPC (SLURM), CI/CD validation, Containers
AI / ML	AI agent workflows (GH-AW, agent contracts), predictive analytics, intelligent forecasting, automated decision support, PINNs
CONSORTIA	Co-maintainer of EC-Earth4 across 30 partner institutes; operational release coordination, integration standards, CI/CD, documentation
MENTORING	Supervision of 2 PhD and 3 Master’s students; organization of training events
OUTPUT	27 peer-reviewed publications; 21 international presentations

PROFESSIONAL EXPERIENCE

2025 – now	Senior Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE	◦ Capacity & Workflow Orchestration: Co-lead for Auto-EC-Earth4; architecting workflow execution DAGs (Autosubmit/Slurm) scheduling 100s of concurrent compute jobs under cluster capacity constraints, eliminating contention bottlenecks and ensuring on-time delivery for operational production cycles. ◦ Predictive Modeling & Anomaly Detection: Designing probabilistic ensemble prediction systems (50+ members); applying statistical signal detection to identify observed anomalies against historical baselines; evaluating forecast skill and uncertainty across operational configurations. ◦ AI Adoption & Agentic Systems: Leading AI adoption across the research group through Climate Commons (CVC-COMMONS); architecting institutional agent contracts (AGENTS.MD), GitHub/GitLab agentic workflows, and schema validation harnesses for automated workflow ingestion and drift detection.

2019 – 2024	Recognized Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE	◦ State Estimation (Data Assimilation): Designed and validated ensemble Kalman filter and nudging schemes for optimal state estimation in high-dimensional systems ($\mathcal{O}(10^6)$ state variables); reduced systematic biases across multi-decadal integrations. ◦ Experimentation at Scale: Built infrastructure for rapid iteration over assimilation parameters and ensemble configurations (50–100 members $\times$ 60-year integrations), enabling systematic sensitivity analysis and hyperparameter tuning.
2015 – 2017	Postdoctoral Scientist, Ocean Dept., MPI-M, Germany
ROLE	◦ Numerical Modeling & Compute Efficiency: Developed dynamical solvers for the ICON modeling framework; designed and optimized numerical algorithms to eliminate runtime bottlenecks, balancing physical accuracy, numerical stability, and compute efficiency.
2011 – 2014	Research Fellow, Applied Math Dept., Univ. of Leeds, UK
ROLE	◦ Large-Scale Numerical Solvers: Developed global finite-difference solvers for multi-scale dynamical systems ($\mathcal{O}(10^6)$ unknowns); designed and developed a novel second-order boundary scheme improving numerical accuracy at domain interfaces (<i>Ocean Modelling</i>, 2014). ◦ Domain Decomposition & Parallelization: Implemented parallel numerical solvers utilizing domain decomposition over complex boundaries, coupling multi-physics processes across spatial scales. ◦ Parameter Sensitivity Analysis: Systematic multi-parameter sensitivity analysis of governing dynamical equations, identifying stability thresholds across physical regimes.
SUM 2008	Risk Analyst (Intern), Raiffeisenbank, Russia
ROLE	◦ Data Analysis: Portfolio risk diagnostics, quantitative metric tracking, and preparation of operational risk reports.
SUM 2007	Junior Statistician, ACNielsen, Russia
ROLE	◦ Statistics: Statistical analysis of retail consumer purchase data and anomaly detection in store audit datasets.

EDUCATION

2008 – 2011	Ph.D. in Applied Mathematics, University of Limerick, Ireland
ROLE	◦ Applied Math: Research in geophysical fluid dynamics and thin-film Stokes flows using asymptotic analysis and numerical modelling (MATLAB, COMSOL). ◦ Teaching: Teaching Assistant for Engineering Mathematics I/II, Vector Analysis, and Mathematical Modeling and Tutor in the Maths Learning Center. ◦ Thesis: “Resonant over-reflection of waves by jets in a rotating ocean.”
2003 – 2008	B.Sc., M.Sc. in Applied Mathematics, Moscow State University, Russia
	◦ Grades: 4.85/5*, 1st Honours ◦ Thesis: “On the stability of the plane flows of visco-plastic fluids”

SELECTED PUBLICATIONS

2025	Mahmood, R et al. (Mar. 2025). “Multi-decadal initialized climate predictions using the EC-Earth3 global climate model”.
2022	Hermanson, L et al. (Apr. 2022). “WMO Global Annual to Decadal Climate Update: A Prediction for 2021–25”. <i>Bulletin of the American Meteorological Society</i> 103 , E1117–E1129. ISSN: 1520-0477.
2022	Acosta Navarro, JC et al. (May 2022). “Added value of assimilating springtime Arctic sea ice concentration in summer-fall climate predictions”. <i>Environmental Research Letters</i> 17 , 064008. ISSN: 1748-9326.
